# Spike / Feasibility — Offline Bangla Dubbing Video Player

Oct 29, 2026 · @Md. Nazmul Hossain

## Verdict

**Feasible on paper; needs one real run to confirm.** The licence check found a fully commercial-safe offline stack. It also found a Bangladeshi Bangla speech corpus that reduces the biggest risk from the gap analysis.

- **Licences:** cleared. Whisper, pyannote 3.1, IndicTrans2 and Indic Parler-TTS all permit commercial use. NLLB-200 and MMS-TTS are excluded (BRD v1.1, NFR-08).
- **Bangladeshi voice:** SUBAK.KO, a 241-hour Bangladeshi Bangla corpus under CC-BY-4.0, makes an accent fine-tune possible. The BRD now treats accent (BR-04b) as phase 2, so this spike only scores the preset voices.
- **Pass conditions follow BRD v1.1:** 4 guaranteed speakers, 95% of lines within 0.5 s, English and Hindi guaranteed. Only E1 now blocks the project outright.
- **Still unproven:** translation naturalness, voice quality, diarization accuracy, speed and sync. These need the prototype run on real hardware.

The prototype code is written and syntax-checked, but it has not been run against the models yet, because they need a GPU machine with internet for the first download.

## Scope, timebox and test setup

**Timebox:** 5 working days, one developer. Stop when the go / no-go rules below can be applied.

**In the spike:** the pre-process pipeline end to end, a minimal PyQt player, measurements.
**Not in the spike:** live mode, final UI, packaging, voice fine-tuning (only its feasibility).

**Test machines**

| Machine | Purpose |
|---|---|
| PC with NVIDIA GPU, 8 GB+ VRAM | Recommended-hardware timings |
| Mac with Apple Silicon, 16 GB | macOS timings |

| Machine | Purpose |
|---|---|
| CPU-only laptop | Minimum-hardware timings |

**Test clips** (5 minutes each)

| Clip | Why |
|---|---|
| English, 1 speaker (lecture) | Baseline quality |
| English, 4 speakers (film or drama) | BR-05 guaranteed speaker count |
| Hindi, 2 speakers | Guaranteed source language (BR-02) |
| Korean, 2 speakers | Best-effort language via English pivot |
| English with loud music | Background separation and its fallback (BR-06) |

**Reviewers:** at least 3 native Bangladeshi Bangla speakers score each output from 1 to 5.

## Licence verification

Checked against model cards and repositories on 2026-10-29.

| Component | Model | Licence found | Commercial | Notes |
|---|---|---|---|---|
| Speech-to-text | faster-whisper | MIT | Yes | Runs Whisper on CTranslate2 |
| Diarization | pyannote speaker-diarization-3.1 | MIT | Yes | Gated: accept conditions and use a Hugging Face token once |
| Translation | IndicTrans2 | MIT | Yes | English ↔ Bengali (ben_Beng); Indic ↔ Indic |
| Translation | NLLB-200 | CC-BY-NC 4.0 | No | Research model; spike comparison only |

| Component | Model | Licence found | Commercial | Notes |
|---|---|---|---|---|
| Bangla TTS | Indic Parler-TTS | Apache 2.0 | Yes | Gated download; 6 named Bengali voices (Arjun, Aditi, Tapan, Rashmi, Arnav, Riya) |
| Bangla TTS | bn-parler-tts | Not confirmed | Check | Community fine-tune on SUBAK.KO; no named speakers |
| Voice data | SUBAK.KO | CC-BY-4.0 | Yes, with attribution | 241 h Bangladeshi Bangla from SUST; 61 native speakers |

**Key findings**

- IndicTrans2 is MIT, but it only translates between English and Indian languages. Korean, Arabic and other sources need a pivot: Whisper's built-in translate-to-English, then IndicTrans2 English → Bangla.
- Indic Parler-TTS's Bengali voices are trained mostly on Indian data. The SUBAK.KO speakers come from 8 divisions and 34 districts of Bangladesh, making it the best public base for a Bangladeshi voice.
- SUBAK.KO is an ASR corpus, so it needs cleaning before TTS fine-tuning; the bn-parler-tts project shows this route works.

## Experiments

Fill the Result column during the spike run.

| # | Experiment | Pass condition (BRD v1.1) | If it fails | Result |
|---|---|---|---|---|
| E1 | End-to-end offline run | 5-min English clip dubs into Bangla with the network off (after model download) | **Rework** to step 1 | |

| # | Experiment | Pass condition (BRD v1.1) | If it fails | Result |
|---|---|---|---|---|
| E2 | Translation naturalness | IndicTrans2 average reviewer score 3.5/5 or higher; local LLM scored for comparison | Use local LLM; rework only if both fail | |
| E3 | Hindi and pivot languages | Hindi clip scores 3.5/5 or higher; Korean via English pivot scores 3/5 or higher | Hindi fail: tune or rework; Korean fail: stays best-effort | |
| E4 | Preset voices | Score recorded for each of the 6 Indic Parler Bengali voices | Not blocking; informs BR-04b phase 2 | |
| E5 | Diarization | 90%+ correct speaker labels on the 4-speaker clip | Tune once; BR-17 covers residual errors; rework if under 80% | |
| E6 | Speed | Pre-process under 15 min per hour of video on recommended hardware (NFR-10); timings for all machines | Live mode moves to version 2 | |
| E7 | Sync | 95% of lines within 0.5 s; no line over 1.0 s late (NFR-05) | Tune cap; BR-15 editor shortens lines | |
| E8 | Background kept | Reviewers confirm music and effects remain audible | Use audio-lowering fallback ( `--no-separation` ) | |
| E9 | Bangladeshi vocabulary | Reviewers rate wording 3.5/5 or higher; list words to add to the glossary (BR-04a) | Build glossary before development | |

## Prototype code

Delivered as **bangla-dub-spike.zip** in the chat. Two scripts cover experiments E1–E8.

| File | What it does |
|---|---|
| `dub_pipeline.py` | Pre-process pipeline: extract → Demucs → faster-whisper → pyannote → IndicTrans2 → Indic Parler-TTS → time-fit → mix. Writes `.bn.wav`, `.bn.srt`, `.bn.mkv` (Bangla + original tracks), `timings.json` and `lines.json` |
| `spike_player.py` | PyQt6 + mpv player; plays the video with the Bangla track, toggles Bangla / original audio and Bangla subtitles |
| `requirements.txt` / `README.md` | Install steps per OS, gated-model sign-up, how to record each experiment |

### How to run

1. Install ffmpeg, libmpv and PyTorch, then `pip install -r requirements.txt`.
2. Accept the gated model pages on Hugging Face and run once with `--hf-token` to download models.
3. Turn the network off and run `python dub_pipeline.py clip.mp4 --offline`.
4. Play with `python spike_player.py clip.mp4`.

**Tested so far:** both scripts compile; mixing, time-stretch, Bangla subtitles and the two-track MKV export were run on a synthetic clip and work. The model stages have not run yet. Expect small library API fixes on the first real run.

### Design choices in the code

- Non-English speech is translated to English by Whisper first, then English → Bangla by IndicTrans2.
- Each new speaker gets the next voice from Arjun, Aditi, Tapan, Rashmi, Arnav, Riya.
- A Bangla line longer than its slot is sped up, capped at 1.35×; lines at the cap are counted in `timings.json` for E7.

# Go / no-go rules

| Outcome | Decision |
|---|---|
| E1 passes and no other experiment hits its rework condition | **Go** to step 4, Development |
| E1 fails after one tuning retry | **Rework:** back to step 1 (see Rework Plan) |
| E2 fails, a local LLM passes | Go, with a local LLM for translation (raises hardware needs) |
| E5 under 80% after tuning | **Rework:** narrow BR-05 further, or ship the fallback release |
| E3, E4, E6, E7, E8 or E9 fail | Go with the BRD v1.1 fallback listed in the experiments table |
| 2 rework loops on the same experiment | Product owner decides: fallback release (single-speaker dub + subtitles) or not feasible |

**Next step:** run the prototype on the three test machines, fill the Results column above, then apply these rules.

# Re-spike for change requests

UAT feedback and other class C change requests (a new technical approach) come back to this step, as set out in Feedback and Change Management. A class D request (new feature) comes here only when it is risky. A re-spike tests one component, not the whole product.

### When a re-spike runs

| Trigger | Example | Re-spike? |
|---|---|---|
| Class C CR approved | UAT F2 below 3.5: replace or post-edit IndicTrans2 with a local LLM | Yes |
| Class D CR with technical risk | Add Arabic as a guaranteed language | Yes |

| Trigger | Example | Re-spike? |
|---|---|---|
| Class C that blocks UAT approval | Voices unacceptable to most testers | Yes, for the current release |
| Class A, B, E or F | Wording, limits, fallbacks, deferrals | No: goes to step 1, 4 or the backlog |

### Re-spike process

1. **Scope:** name the CR, the component, and the UAT or QA evidence that triggered it.
2. **Timebox:** 3 working days for one component; 5 if it adds a model.
3. **Candidates:** at most 3 options, each checked for a commercial licence and fully offline use before testing.
4. **Experiment:** reuse the matching experiment (E1–E9) with the same test clips, plus the UAT clips where the problem appeared.
5. **Pass condition:** must meet the original experiment's pass condition **and** fix the UAT complaint, for example F2 at 3.5 or higher with the same reviewers.
6. **No regressions:** E1 (end-to-end offline) and E6 (speed) re-run with the new component.
7. **Record:** add a row to the re-spike log below and to the CR.

### Outcomes

| Result | Next step |
|---|---|
| A candidate passes | Update Gap Analysis, then step 4, Development, with the CR ID |
| No candidate passes, but a BRD fallback exists | Reclassify the CR as class E and apply the fallback |
| No candidate passes, no fallback | Defer the CR (class F); release stays as UAT approved it, or follow the Rework Plan if it blocks release |

### Re-spike log

| CR | Date | Component | Candidates tested | Winning option | Experiment results | Decision |
|---|---|---|---|---|---|---|
| | | | | | | |

## Sources

- faster-whisper (GitHub)
- pyannote speaker-diarization-3.1 model card
- IndicTrans2 (GitHub)
- IndicTrans2 en-indic-dist-200M model card
- NLLB-200 (fairseq README)
- NLLB-200 distilled 600M model card
- Indic Parler-TTS model card
- bn-parler-tts model card
- SUBAK.KO dataset card
- RegSpeech12 paper (SUBAK.KO speaker regions)

## Change log

| Version | Date | Change |
|---|---|---|
| 1.2 | 2026-10-29 | Added re-spike process for change requests from UAT and Feedback and Change Management |
| 1.1 | 2026-10-29 | Aligned with BRD v1.1 and the Rework Plan: 4-speaker test clip, Hindi as guaranteed language, new E9 for Bangladeshi vocabulary |
| 1.1 | 2026-10-29 | Each experiment now names its fallback; only E1 (and E5 under 80%) forces rework |
| 1.0 | 2026-10-29 | First version |